

SQL INTRODUCTION

◆ What is Data:-

- Data refers to raw facts, numbers, or symbols that **represent information.**
- For example: **names, marks, prices, dates.**

◆ Types of Data:-

- **Structured Data:** Organized in **rows and columns like table format.**
Example: **Excel sheet, RDBMS.**
- **Semi-Structured Data:** Not stored in a **fixed format.**
Example: **JSON, XML.**
- **Unstructured Data:** **No specific format.**
Example: **Images, videos, emails, audio files.**

◆ RDBMS:-

- RDBMS stands for **Relational Database Management System.**
- Stores data in tables is called **relations.**
- Allows data to be related across tables using keys (like **primary key & foreign key**)
- Uses SQL (Structured Query Language) to **manage and query data.**
- Example of RDBMS :- **MySQL, Oracle, Microsoft SQL Server, PostgreSQL**

◆ What is a Database?

- A database is a structured collection of related **data stored** and accessed electronically.
- It allows data **create, storage, retrieval, and manipulation.**

◆ Database History:-

- 1960s: Hierarchical and Network Models.
- 1970s: Relational Model introduced by Dr. E.F. Codd (basis for SQL).
- 1980s: SQL became a standard and widely used.
- Today: RDBMS like MySQL, Oracle, PostgreSQL, etc., dominate.

◆ What is SQL?

- SQL stands for **Structural Query Language**.
- **Designed for managing data** in a relational database management system(**RDBMS**).
- It is pronounced as S-Q-L or **See-Qwell**.
- SQL is a database language, it is used for database **creation**, **deletion**, **fetching rows**, and **modifying rows**, etc.
- SQL is based on relational algebra and tuple relational calculus.

◆ Why SQL is Required?:-

- **Create** new databases, table and views.
- **Insert** records in a database.
- **Update** records in a database.
- **Delete** records from a database.
- **Retrieve** data from a database.

◆ SQL Syntax:-

- SQL is not case sensitive.
- SQL keywords are written in uppercase.
- We can place a single SQL statement on one or multiple text lines.
- SQL statements are started with any of the SQL commands/keywords like SELECT, INSERT, UPDATE, DELETE, ALTER, DROP etc. and the statement ends with a semicolon(;).

◆ Important SQL Commands:-

